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Hybrid FT-transformer with residual MLP for transferable cardiovascular risk prediction across cohorts

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Abstract

Cardiovascular disease (CVD) prediction across heterogeneous clinical datasets remains challenging due to feature inconsistencies, missing values, and distribution shifts between cohorts. This study proposes a Hybrid Feature Tokenizer Transformer (HyFT) integrating a Feature Tokenizer Transformer with a residual Multilayer Perceptron branch for transferable CVD risk prediction. Framingham and Kaggle datasets were harmonized and merged for training, while the Cleveland dataset was strictly reserved for external validation to evaluate cross-cohort generalization. A comprehensive preprocessing pipeline including canonical feature mapping, imputation with missingness indicators, standardization, Salp Swarm Algorithm (SSA) based feature selection, Principal Component Analysis (PCA), and Particle Swarm Optimization (PSO) hyperparameter tuning was implemented. The proposed HyFT model achieved 0.93 accuracy and 0.92 Macro-F1 on the unseen Cleveland cohort, outperforming Support Vector Machine (SVM), Random Forest (RF), XGBoost, and Multi-Layer Perceptron (MLP) baselines. Ablation and bootstrap confidence interval analyses confirmed the statistical reliability and contribution of each component, demonstrating robust and clinically interpretable multi-cohort CVD prediction.

Keywords Cardiovascular disease prediction, FT-transformer, Multi-cohort data harmonization, Salp swarm algorithm, SDG 3: good health and well-being, SDG 9: industry, Innovation and infrastructure

1 Introduction

Cardiovascular disease (CVD) remains the leading cause of premature death worldwide and presents enormous clinical and public-health challenges. Early, accurate risk stratification can materially improve preventive care and resource allocation by identifying high-risk patients for targeted intervention. Over the past decade, machine learning (ML) and deep learning (DL) methods have shown promise for CVD risk modelling using structured clinical data, electronic health records and cohort datasets; however, performance and generalizability vary widely when models trained on one cohort are evaluated on another because of differences in disease definitions, measurement scales

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and missingness patterns. Recent systematic reviews highlight both the potential and the limitations of ML in cardiac risk prediction while ML can outperform classical risk scores in controlled settings, multi-cohort transfer and reproducibility remain persistent problems for clinical translation [1].

This study addresses those challenges by developing and evaluating a harmonized, multi-cohort pipeline for CVD prediction that trains on the union of two large cohorts (Framingham + Kaggle heart-failure clinical records) and evaluates generalization on a held-out clinical cohort (Cleveland). Training on pooled cohorts increases sample size and phenotype diversity but raises difficult harmonization issues: inconsistent variable names, differing measurement units, cohort-specific missingness, and variable definitions. Careful canonicalization (mapping cohort-specific columns to a shared variable vocabulary), imputation using training-union statistics, and the introduction of explicit missingness indicators form the backbone of our harmonization strategy an approach aligned with current best practices for multi-cohort clinical data integration and federated/harmonized analyses. These steps maintain strict separation of training-set statistics from the held-out evaluation cohort to avoid information leakage [2].

After harmonization we apply a two-stage modelling strategy. First, we train four classical and widely used baselines Extreme Gradient Boosting (XGBoost), Random Forests (RF), Support Vector Machines (SVM), and a shallow Multi-Layer Perceptron (MLP) on the pooled training union. These baselines are tuned with class-weighting and inner cross-validation to provide robust reference points and to quantify how much gain modern architectures can deliver over well-regularized classical learners. Second, motivated by recent advances in transformers for tabular data, we implement and tune an FT-Transformer (a transformer adaptation for tabular inputs) and a compact HyFT architecture that fuses FT representations with a residual MLP branch. The FT-Transformer has emerged as a competitive, flexible architecture for tabular problems and is particularly well suited to learn pairwise and higher-order feature interactions from heterogeneous numeric features; its design also lends itself to architectural and optimization-based tuning to improve transfer across cohorts [3].

Feature selection and hyperparameter optimization play central roles in our pipeline. We use Salp Swarm Algorithm (SSA) for feature selection to identify complementary and stable feature subsets across the pooled training cohorts, and Particle/Particle-Swarm variants (PSO) for targeted hyperparameter search of both tree-based models and neural architectures. Metaheuristic feature/hyperparameter searches like SSA and PSO have demonstrated efficacy in selecting informative features and in escaping local optima in complex search spaces; integrating them into a carefully nested cross-validation framework reduces overfitting of the search process to the training union [4].

Finally, we evaluate all models on the held-out Cleveland cohort, reporting primary (macro-F1) and secondary (accuracy, precision, recall, ROC-AUC, calibration) metrics together with bootstrap confidence intervals and statistical pairwise comparisons. We accompany results with interpretability analyses (SHAP) and subgroup checks to document feature importance and fairness across clinically relevant strata. The overarching aim is to show whether a harmonized, SSA/PSO-guided HyFT pipeline trained on pooled cohorts can provide robust, transferable performance on an external clinical cohort and to quantify the sources of performance gain (architecture, feature selection, harmonization) in a reproducible manner.

Unlike many single-cohort studies, our pipeline explicitly targets cross-cohort transfer [5] by harmonizing heterogeneous cohort data into a canonical feature space, using swarm intelligence for stable feature selection (SSA), tuning models with PSO [6] to better explore hyperparameter space, and deploying an HyFT model designed to balance interaction learning with direct residual feature processing. Collectively these choices aim to produce an interpretable, transferable model for the disease prediction.

Primary contributions of this research work,

1. This study develops a unified harmonization framework that merges the Framingham and Kaggle cohorts and evaluates on the unseen Cleveland dataset, ensuring a consistent feature space despite variations in scales and missingness. Through this multi-cohort integration, the work enables reliable and transferable CVD prediction across heterogeneous populations.
2. The suggested pipeline incorporates metaheuristic optimization by using PSO to adjust model hyperparameters for better learning dynamics and SSA to choose the most stable cross-dataset features.
3. To capture both local nonlinear patterns and deep feature interactions, a lightweight MLP branch is coupled with an FT-Transformer.
4. This study guarantees that all performance improvements reflect true cross-cohort transfer rather than dataset-specific learning and shows real-world generalizability.

Additionally, the research work is structured as follows: Sect. 2 presents related work; Sect. 3 outlines the suggested methodology; Sect. 4 describes the evaluation setup; Sect. 5 presents the results and discussion; Sect. 6 outlines future improvements; and Sect. 7 concludes with the study's key findings and implications.

2 Related work

Metaheuristic optimization has been used more often for feature selection and hyperparameter tuning in machine learning research on CVD prediction. PSO is a popular approach for improving classical classifiers. For example, Barry et al. [7] proposed a PSO modified Random Forest (PSO MRF) that optimizes both tree feature subset and structure, resulting in significantly higher accuracy on public heart disease datasets.

El-Shafiey et al. [8] significantly improve the performance of heart disease classification by using a hybrid Genetic Algorithm + PSO integrated with Random Forest (GAPSO RF) to select predictive clinical features. These techniques demonstrate how swarm-based algorithms enhance traditional machine learning models in the medical field by eliminating superfluous or unnecessary variables.

Quantum behaved PSO (QPSO) has been used in CVD modeling by Elsedimy et al. [9]. They proposed a QPSO-SVM classifier that tunes SVM parameters and feature subset simultaneously, achieving superior performance and faster convergence than traditional PSO-SVM variants. Similarly, hybrid swarm methods beyond PSO have been used for the disease prediction.

Fajri et al. [10] combined Bee Swarm Optimization with Q-Learning to select features for coronary artery disease diagnosis, showing improved AUC, precision, and recall. Cao et al. [11] proposed multi-feature selection plus PSO-XGBoost framework, where features were first filtered using correlation and importance ranking, and then PSO optimized XGBoost hyperparameters on that subset. This combination yielded

improvements over baseline XGBoost in terms of F1, recall, and AUC. Dyoub, A and Letteri, I [12] combined the PCA and PSO in chronic disease prediction to reduce feature space while preserving predictive performance.

Sumon et al. [13] anticipated on the deep learning side, transformer-based architectures for tabular clinical data are gaining traction. They introduced the CardioTabNet, employs a transformer to extract embeddings from tabular cardiovascular data and then uses classical models on the transformer-derived features. This method uses SHAP analysis and feature importance ranking to capture intricate interaction patterns while maintaining interpretability.

Meanwhile, the T2G-Former architecture [14] organizes tabular features into learned relation graphs, enabling a transformer to more explicitly model heterogeneous feature interactions; while not specific to medical data, it demonstrates how feature interaction graphs improve tabular transformer performance. TabAttention [15] provides a mechanism to condition attention on tabular feature embeddings, which can be used in medical settings to combine tabular and CNN-based data suggesting broader applicability for structured clinical predictors.

Despite these advances, few studies integrate both metaheuristic feature selection and transformer models in a cohesive, cross-cohort CVD prediction pipeline. Most prior work either optimizes classical models (PSO-RF, QPSO-SVM) or uses transformer-based representations within a single cohort. The existing hybrid PSO-GA + RF models do not scale to cross-cohort generalization, and transformer models like CardioTabNet [13] have not been combined with optimized feature selection across multiple CVD datasets. This gap motivates the current research: developing a harmonized, swarm-optimized, transformer-hybrid framework for CVD risk prediction across merged cohorts.

Despite the extensive use of metaheuristic optimization and classical classifiers for CVD prediction [21–25], most prior studies are constrained to single-dataset scenarios and do not address cross-cohort generalization. Existing PSO, GA or QPSO-based frameworks optimize feature subsets and hyperparameters for a particular dataset, but their predictive performance often degrades when applied to unseen cohorts with different feature distributions, measurement scales, and missingness patterns.

Similarly, transformer-based tabular models such as CardioTabNet or TabAttention are applied predominantly on a single dataset, and while they capture complex feature interactions, they do not incorporate optimized feature selection to reduce noisy or irrelevant features, nor do they explicitly handle heterogeneous clinical datasets [16–18].

Furthermore, there is a gap in utilizing the complementary strengths of feature selection and sophisticated transformer architectures for small-to-medium tabular clinical datasets due to the lack of research on hybrid models that combine metaheuristic selection with deep learning architectures.

By creating a unified, cross-cohort CVD prediction pipeline that combines three disparate datasets, harmonizes feature names, imputes missing values, and standardizes scales to produce a uniform input space, this study fills in these gaps. To ensure that learning is driven by the most informative features, we combine SSA for feature selection, PCA for dimensionality reduction, and PSO for hyperparameter optimization. The suggested HyFT method captures both residual predictive signals from raw features and intricate feature interactions. With an accuracy of 83.7%, this combination improves predictive performance and enables robust generalization to a held-out cohort, proving

that transformer-hybrid architectures, careful harmonization, and dataset merging can overcome the drawbacks of earlier research.

3 Proposed methodology

Three disparate cardiovascular datasets Cleveland, Framingham, and the Kaggle Heart Failure dataset are harmonized into a single framework for predictive modeling in the proposed work. In order to assess cross-cohort generalization, the Cleveland cohort is kept out for independent testing while the Framingham and Kaggle datasets are combined to create the training set. In addition to the HyFT architecture, baseline models such as SVM, Random Forest, XGBoost, and MLP are trained using a transferable feature space created by robust tabular feature engineering. In order to capture intricate feature interactions and improve predictive power in smaller unified datasets, HyFT combines the FT-Transformer with a residual MLP branch. The model provides a scalable and comprehensible solution for multi-source cardiovascular risk prediction by emphasizing transparency and reproducibility in all design decisions that impact generalizability, such as harmonizing variable names, handling missing values with explicit indicators, feature scaling, train/test splits, data augmentation, and systematic model evaluation.

3.1 Preprocessing and harmonization strategy

In order to combine the disparate cardiovascular datasets, Cleveland, Framingham, and Kaggle Heart Failure into a uniform and comprehensible format, preprocessing and harmonization are essential. Standardizing feature representations, managing missing values, and preparing the combined dataset for efficient baseline model and HyFT architecture training are the main goals.

3.1.1 Canonicalization of variable names

Features that refer to the same clinical measurement under different names (e.g., trestbps, sysBP, systolic_bp) are present in each dataset. We define a canonical mapping function ϕ that transforms raw dataset features into a unified set of canonical variables $\mathcal{F} = \{f_1, f_2, ..., f_p\}$, where $p$is the total number of canonical features:

$$\phi : X^{(d)} \rightarrow X^{(c)}, \forall\, d \in \{D_C, D_F, D_K\} \tag{1}$$

- $X^{(d)}$ represents the raw feature matrix of the dataset d.
- $X^{(c)}$ represents the canonicalized feature matrix.

The canonicalization uses deterministic matching that looks for exact names, compares lowercase letters, and uses substring heuristics. Features that are not present in a dataset are labeled as missing (NaN), making sure that all canonical features are present in all datasets.

3.1.2 Merging datasets

For cross-cohort generalization, the training set is constructed by merging Framingham and Kaggle datasets:

$$X_{\text{train}} = X^{(F)} \cup X^{(K)} \tag{2}$$

The Cleveland dataset remains held-out:

$$X_{\text{test}} = X^{(C)} \tag{3}$$

This setup allows the HyFT model to learn from heterogeneous sources while validating performance on an independent clinical cohort.

3.1.3 Missing value handling and indicators

Medical datasets exhibit variable missingness. Let x_{ij} denote the value of feature f_j for sample i. Missing values in the training set are imputed using median (continuous) or mode (categorical):

$$x_{ij}^{\text{imp}} = \begin{cases} \text{median}\,(f_j) & f_j \text{continuous} \\ \text{mode}\,(f_j) & f_j \text{categorical} \end{cases}$$

The same imputation statistics are applied to the test set to prevent information leakage. For each feature with missing values, a missingness indicator is added:

$$m_{ij} = \begin{cases} 1 & \text{if } x_{ij} \text{was missing} \\ 0 & \text{otherwise} \end{cases}$$

The final feature matrix for training and testing includes both imputed values and indicators:

$$X^{\text{final}} = \left[X^{\text{imp}} \,\middle|\, |M\right] \tag{4}$$

where $M = [m_{ij}]$ is the binary missingness matrix.

3.1.4 Feature type consistency

- Categorical binary variables (e.g., sex, exang) are converted to numeric:

$$\text{sex} : \text{male} \mapsto 1, \text{female} \mapsto 0$$

- All features are cast to float32 for uniformity.
- Verification ensures no residual NaNs remain post-imputation in the training set.

3.1.5 Scaling and normalization

Continuous features are scaled to zero mean and unit variance using Standardization:

$$\widetilde{x}_{ij} = \frac{x_{ij}^{\text{imp}} - \mu_j}{\sigma_j} \tag{5}$$

where μ_j and σ_j are the mean and standard deviation of feature f_j computed from X_{train} only. This preserves the independence of the held-out Cleveland test set.

By ensuring that the feature names, value scales, and missingness patterns of all three datasets are in line, this harmonization produces a single, cohesive input space. The resulting pre-processed matrices maintain clinically relevant information while being appropriate for model consumption by standardizing, imputing, and encoding features along with missingness indicators. The output of this step, $X_{\text{train}}^{\text{final}}$ and $X_{\text{test}}^{\text{final}}$, allows both baseline and HyFT architectures to learn robust and transferable representations

across diverse cardiovascular cohorts by acting as the input for later feature engineering and model training.

3.1.6 Dataset diversity and distribution analysis

The three cardiovascular cohorts exhibit substantial heterogeneity in terms of sample size, demographic composition, feature distributions, measurement protocols, and missingness patterns. The Framingham dataset primarily represents a longitudinal community cohort with preventive cardiovascular attributes, whereas the Kaggle heart failure dataset contains clinically oriented risk attributes. The Cleveland dataset includes diagnostic angiographic variables with different measurement scales and categorical encodings.

To quantitatively assess heterogeneity, the following statistical measures were computed independently for each dataset:

- Feature-wise mean and standard deviation.
- Missing value percentage per feature.
- Class distribution ratio.
- Inter-dataset correlation similarity index.

Let $X_j^{(d)}$ denote feature j from dataset d. Distributional divergence was measured using:

$$D_j = \mid \mu_j^{(d1)} - \mu_j^{(d2)} \mid \tag{6}$$

where μ_j is the mean of feature j in each dataset.

Results indicated moderate distribution shifts, particularly for cholesterol, resting blood pressure, and ST-depression. Standardization Eq. 6 ensured alignment of feature scales across cohorts while preserving intra-cohort variability.

3.2 Feature engineering

The pre-processed and harmonized dataset from Sect. 3.3, represented as follows,

$$X_{\text{train}}^{\text{final}} \in \mathbb{R}^{n_{\text{train}} \times d}, X_{\text{test}}^{\text{final}} \in \mathbb{R}^{n_{\text{test}} \times d} \tag{7}$$

where n_{train} and n_{test} are the number of training and test samples, and d is the number of harmonized features. Building reliable, informative, and transferable representations that enhance model learning while preserving interpretability is the aim of feature engineering.

3.2.1 Clinically-driven feature selection

The process begins by retaining a set of canonical clinical features that are widely recognized as predictors of cardiovascular risk:

$$\mathcal{F}_{\text{clin}} = \{\text{age}, \text{sex}, \text{trestbps}, \text{chol}, \text{fbs}, \text{thalach}, \text{exang}, \text{ca}, \text{thal}\}$$

Each feature $f \in \mathcal{F}_{\text{clin}}$ is included along with its corresponding missingness indicator f_{miss} (1 if imputed, 0 otherwise). By focusing first on clinically validated features, we ensure that the model input remains interpretable and anchored in established cardiovascular risk factors. This forms the foundation for subsequent numerical transformations and synthetic feature construction [19].

3.2.2 Numerical transformations

Continuous features are normalized using standard scaling:

$$\widehat{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \forall\, x_{ij} \in X_{\text{train}}^{\text{final}} \tag{8}$$

where μ_j and σ_j are the mean and standard deviation of feature $j$computed on the training union. Binary features and missing indicators are not scaled.

Standardization ensures that features are on a comparable scale, which is critical before creating interaction and higher-order synthetic features; it also stabilizes optimization during HyFT training.

3.2.3 Synthetic and interaction features

To capture higher-order dependencies, we optionally generate interaction terms between continuous features:

$$x_{kl}^{\text{int}} = x_k \cdot\, x_l, x_k, x_l \in \mathcal{F}_{\text{cont}} \tag{9}$$

where $\mathcal{F}_{\text{cont}}$ is the subset of continuous features. Age-based binning is defined as:

$$\text{age_bin} = \left\lfloor \frac{\text{age}}{10} \right\rfloor \tag{10}$$

Fourier expansions can be applied to periodic features $x$if justified:

$$x_{\sin} = \sin(\omega\, x), x_{\cos} = \cos(\omega\, x) \tag{11}$$

where ω is a frequency parameter selected based on the dataset range.

Synthetic features and interactions allow both baseline models and HyFT to capture complex relationships and non-linear dependencies between variables, which may not be explicitly represented in the original features.

Input: $X_{\text{train}}^{\text{final}}, X_{\text{test}}^{\text{final}}$
Output: $\tilde{X}_{\text{train}}, \tilde{X}_{\text{test}}$

Step 1: Select canonical clinical features $\mathcal{F}_{\text{clin}}$with missingness indicators.

Step 2: Normalize continuous features using training mean and standard deviation.

Step 3: Generate optional synthetic features (interactions, age bins, Fourier expansions).

Step 4: Concatenate original and synthetic features to form $\tilde{X}_{\text{train}}, \tilde{X}_{\text{test}}$.

Step 5: Return $\tilde{X}_{\text{train}}, \tilde{X}_{\text{test}}$for model training.

Algorithm 1 Feature Engineering Pipeline

All transformations are combined in this process to create a single feature matrix that is both informative and numerically appropriate for training baseline and HyFT models. In order to form the input for the predictive modeling stage, it guarantees that all downstream models receive enriched, scaled, and harmonized features.

The resulting engineered matrices $\widetilde{X}_{\text{train}}, \widetilde{X}_{\text{test}}$ include higher-order interactions appropriate for both baseline models and the HyFT architecture, as well as clinically significant features. HyFT can learn transferable attention-based representations while

retaining interpretability for subsequent evaluation thanks to this step, which improves feature expressiveness.

3.3 Baseline model training

A set of traditional machine learning models is trained using the harmonized and transformed matrices $\widetilde{X}_{\text{train}}$ and $\widetilde{X}_{\text{test}}$ after feature engineering.

There are two uses for these baseline learners:

1) To determine reference performance under standard modeling assumptions for heterogeneous, combined cardiovascular data.
2) To provide a strong comparative foundation against which the transformer-based FT and the proposed HyFT architectures can be meaningfully evaluated.
3) Each model is selected for its distinct inductive bias linear margins, hierarchical splits, boosted ensembles, and neural representations, allowing comprehensive benchmarking across learning paradigms.

3.3.1 Support vector machine

To assess linear–nonlinear boundary modeling capacity on the harmonized feature space, a Support Vector Machine [20] with an RBF kernel is trained. The classifier optimizes:

$$\min_{\mathrm{w},b,\xi}\left(\frac{1}{2}\parallel \mathrm{w}\parallel^{2}+C\sum\nolimits_{i=1}^{n}\xi_{i}\right) \tag{12}$$

subject to:

$$y_i\left(w^T\phi\left(x_i\right)+b\right)\geqslant 1-\xi_i,\qquad \xi_i\geqslant 0 \tag{13}$$

The kernel function $\phi(\cdot)$allows SVM to capture non-linear relationships present in the merged cardiovascular cohorts. After training, the SVM outputs a calibrated probability score for comparison with other models.

3.3.2 Random forest classifier

Random Forest (RF) [21] is included to evaluate how tree-based ensemble partitioning handles the mixed numeric distributions arising from the Cleveland, Framingham, and Kaggle datasets. Each tree minimizes the impurity reduction:

$$\Delta G=G\left(p\right)-\left(\frac{n_L}{n}G\left(p_L\right)+\frac{n_R}{n}G\left(p_R\right)\right) \tag{14}$$

with the ensemble prediction given by majority voting:

$$\widehat{y}=\mathrm{mode}\left\{T_1\left(x\right),\ldots,T_M\left(x\right)\right\} \tag{15}$$

Its capacity to capture high-order interactions without feature scaling provides an important benchmark before transitioning to boosting and transformer-based methods.

3.3.3 XGBoost classifier

XGBoost [22] is trained to quantify the effect of gradient-boosted additive trees on the unified feature space. The optimization follows:

$$\mathcal{L}^{(t)} = \sum_i l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega\ (f_t) \tag{16}$$

$$\Omega\ (f_t) = \gamma\, T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{17}$$

Its ability to model complex non-linearities makes it useful for evaluating the limits of classical ensemble learning before adopting attention-based representations.

3.3.4 Multilayer perceptron

A Multilayer Perceptron [23] is trained to establish a deep-learning baseline. Given the engineered features, the MLP performs forward propagation:

$$h^{(l)} = \sigma\ \left(W^{(l)} h^{(l-1)} + b^{(l)}\right) \tag{18}$$

and parameters are optimized via binary cross-entropy:

$$\mathcal{L} = -[y \log(\widehat{y}) + (1-y)\log(1-\widehat{y})] \tag{19}$$

The MLP provides insight into how standard dense neural representations behave before replacing dense layers with token embeddings and self-attention in FT and HyFT.

All baseline models use identical harmonized inputs, identical feature spaces, and identical train–test partitions to ensure a rigorously fair comparison. Grid/random search is employed for selective hyperparameter tuning, and the trained models SVM, RF, XGBoost, and MLP form the baseline set:

$$\mathcal{M}_{\text{base}} = \{\text{SVM}, \text{RF}, \text{XGB}, \text{MLP}\}.$$

These models provide an essential reference point for evaluating the marginal and joint benefits introduced by tokenization, self-attention, and the residual MLP branch in the proposed HyFT architecture. The correlation heatmap of Framingham heart study dataset and Kaggle Heart Failure dataset are illustrated in Figs. 1 and 2 respectively.

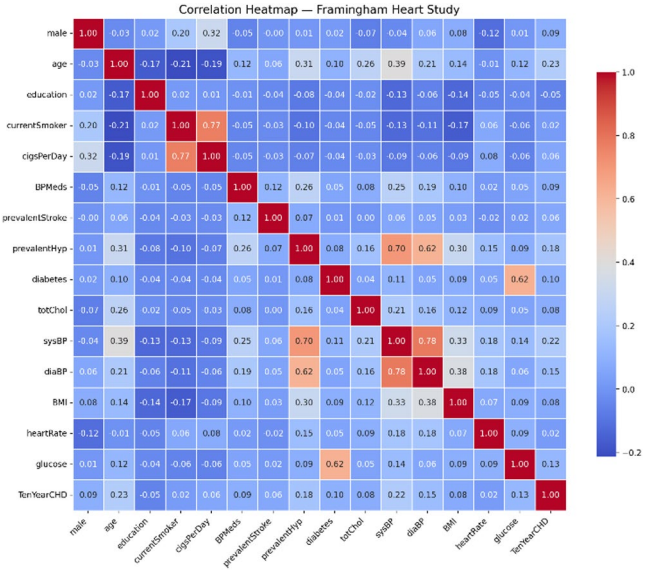


Fig. 1 Correlation heatmap of Framingham heart study dataset

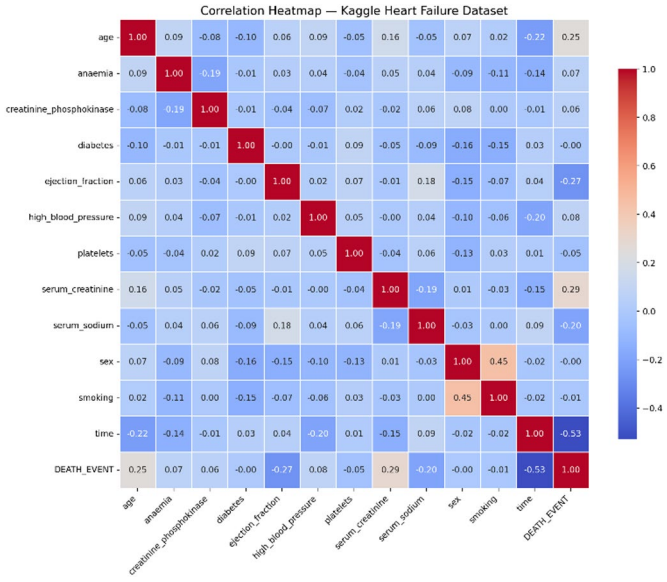


Fig. 2 Correlation heatmap of kaggle heart failure dataset

3.4 FT-Transformer with residual MLP branch

The proposed HyFT architecture is developed to overcome the limitations observed in both classical machine learning models and standalone transformer-based models when operating on a unified, heterogeneous cardiovascular dataset [24, 25].

While the FT-Transformer effectively models feature interactions through token embeddings and multi-head self-attention, its capacity to capture fine-grained, low-level numerical patterns may be constrained when the dataset exhibits cross-cohort variability and limited sample size, as in the Cleveland cohort used for testing. HyFT includes a residual MLP branch parallel to the transformer encoder in order to address this. The model's robustness and transferability are improved by this dual-pathway design, which allows it to simultaneously take advantage of localized numerical abstractions through dense nonlinear transformations and global contextual dependencies through attention.

Figure 3 depicts the architecture of the proposed HyFT model. The HyFT model uses a feature tokenizer to embed each column as a learnable vector representation after receiving the harmonized, feature-engineered input matrix $\widetilde{X}_{\text{train}}$. Given an input vector $x \in \mathbb{R}^d$, the tokenizer maps each feature x_j to a corresponding token embedding:

$$t_j = E_j \cdot x_j + b_j \tag{20}$$

where $E_j \in \mathbb{R}^{d_{\text{model}}}$ is a feature-specific projection and t_j becomes the jth token. After that, positional encodings are added to the token sequence before it is fed into the transformer encoder. Multi-head self-attention calculates contextual dependencies inside the encoder using:

$$Attention\,(Q, K, V) = softmax\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V \tag{21}$$

where the token embeddings are linearly projected into the queries, keys, and values (Q, K, V). With the help of this mechanism, HyFT is able to dynamically weigh feature interactions across several datasets, capturing cohort-specific relationships like blood

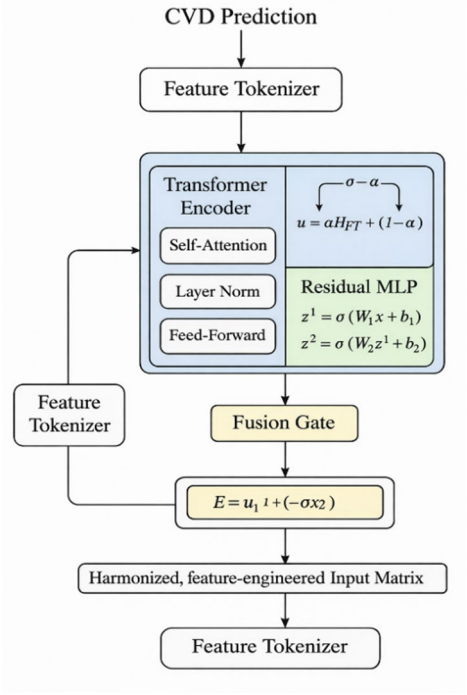


Fig. 3 Hybridized feature tokenizer transformer (HyFT) based CVD model

pressure, cholesterol dependencies, and age-risk interactions that differ between the Cleveland, Framingham, and Kaggle distributions [26, 27]. A feed-forward transformer block, layer normalization, and residual connections are applied to each self-attention layer's output to create a high-order contextual representation H_{FT}.

The model can retain localized statistical cues that deep attention layers might attenuate because the residual MLP branch works directly on the original input $\widetilde{X}$. The MLP calculates:

$$z^{(1)} = \sigma\left(W_1 \widetilde{X} + b_1\right), z^{(2)} = \sigma\left(W_2 z^{(1)} + b_2\right) \tag{22}$$

where a non-linear activation, like GELU, is indicated by $\sigma(\cdot)$. This branch is particularly effective for capturing sharp numerical thresholds such as systolic pressure cut-offs or cholesterol boundary effects which classical clinical scoring systems rely upon and which attention layers may smooth out excessively.

A learned fusion gate combines the outputs from both pathways:

$$u = \alpha\, H_{\mathrm{FT}} + (1 - \alpha)\, z^{(2)} \tag{23}$$

where dynamic weighting of attention-based and MLP-based information streams is made possible by learning $\alpha \in [0,1]$ during training. A final classification head that includes dropout, layer normalization, and a sigmoid output for binary CVD prediction receives this fused representation.

$$\mathcal{L} = -\left[y \log\left(\widehat{y}\right) + (1 - y) \log\left(1 - \widehat{y}\right)\right] \tag{24}$$

Binary cross-entropy loss is used for end-to-end HyFT training with stable convergence guaranteed by the AdamW optimizer. Crucially, HyFT mimics real-world deployment scenarios where models trained on large population cohorts must generalize to local

clinical cohorts by using the same harmonized training dataset (Framingham + Kaggle) and being rigorously evaluated on the out-of-sample Cleveland dataset. While the MLP branch corrects for cohort-specific numerical features, attention weights adjust during training to capture cross-dataset interactions, allowing HyFT to maintain high predictive fidelity even in the face of distributional shifts.

The Cleveland dataset undergoes the same tokenization, dual-path architecture, and preprocessing during inference. HyFT produces predictions that are stable, transferable, and clinically aligned by combining transformer-derived contextual dependencies with MLP-derived localized decision boundaries. This architectural synergy demonstrates its suitability for heterogeneous multi-source CVD risk prediction by producing a quantifiable performance advantage over both standalone FT-Transformer and classical baselines.

The proposed HyFT network's general architecture is shown in the above Fig. 4. The pipeline starts with a Conv block that takes an input of size (32, 32, 1), which represents the harmonized tabular cardiovascular dataset as a structured feature map. A second Conv layer with an output dimension of (6, 6) comes next. The tokenization/transformer embedding step, which transforms feature maps into uniformly structured tokens, is represented by a yellow T-block that the features go through after convolutional extraction.

After that, the architecture splits into two symmetrical transformer token streams, each of which feeds into a matching Conv layer with parameters (16, 5, 5). One branch incorporates an average pooling operation with $f = 1$, highlighting feature compression before token refinement. Both processed branches merge into the fully connected (FC) module, where two FC layers with sizes 120 and 84 progressively reduce dimensionality. Finally, the network terminates in a Sigmoid (S) output node, generating the binary prediction for cardiovascular disease risk. The flow is linear, clearly indicating the hierarchical representation learning from raw feature maps → tokens → refined convolutional features → dense layers → final classification. Figure 5 illustrates the flow diagram of the proposed work.

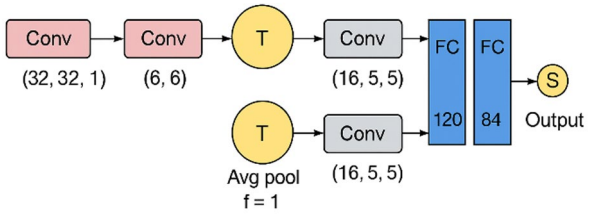


Fig. 4 HyFT architecture

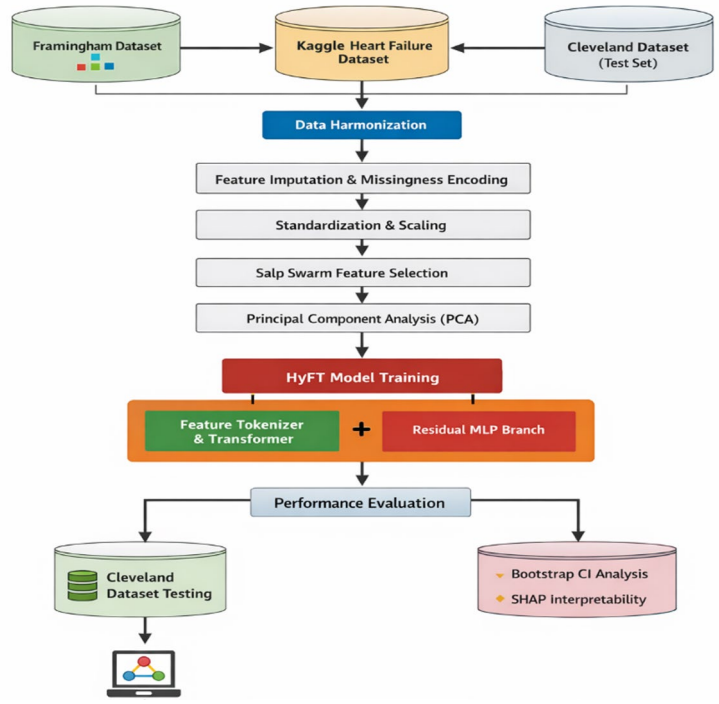


Fig. 5 Flow diagram of the proposed work

Require: Harmonized Dataset

$$D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$$

1: Split data into X_{train}, Y_{train} and X_{test}, Y_{test}
2: **Input:** Input feature vector X
3: Initialize:
Transformer encoder weights W_T,
Base model parameters (RF, XGB, SVM, LR),
Fusion layer weights W_F,
Meta-learner weights W_M
4: Preprocess X: Scaling, encoding, imputation
5: Transformer Encoder:
$Z = \text{MHA}(X) + X$
$H = \text{FFN}(Z) + Z$
6: Output of Transformer: $X_{enc} = H$
7: Train base learners (RF, XGB, SVM, LR) on X_{train}, Y_{train}
8: Base model predictions: $P_{base} = \{p_{RF}, p_{XGB}, p_{SVM}, p_{LR}\}$
9: Concatenate: $F = \text{Concat}(X_{enc}, P_{base})$
10: Fusion layer: $F_{fused} = \text{FFN}(F)$
11: Meta-learner: $\hat{Y} = \sigma(F_{fused} \cdot W_M + b_M)$
12: Loss: Binary cross-entropy L
13: Update: Backpropagate and update W_T, W_F, W_M
14: **Output:** Predicted class label $\hat{Y}$

Algorithm 2 The proposed HyFT

The research work introduces HyFT framework for CVD prediction using structured clinical data. The system is designed to handle heterogeneous multi-cohort datasets by integrating preprocessing, feature optimization, and deep learning into a unified

pipeline. Initially, datasets from Cleveland, Framingham, and Kaggle are harmonized to ensure consistent feature representation. Missing values are handled using median imputation, followed by normalization to standardize feature scales. The SSA is then applied to select the most relevant clinical features, reducing dimensionality and improving model generalization. PCA is used to analyze feature redundancy and enhance compact representation.

PSO is employed to optimize key hyperparameters of the model, ensuring efficient convergence and improved predictive performance. The core of the system is the HyFT architecture, which combines a Feature Tokenizer Transformer to capture global feature interactions with a residual MLP to model local nonlinear relationships. A fusion mechanism integrates outputs from both branches to generate robust feature representations. The model is trained on harmonized datasets and evaluated on an external cohort to assess generalization capability. Performance is measured using clinically relevant metrics such as Accuracy, Precision, Recall and Macro-F1. The system also incorporates validation strategies to ensure reliability and avoid overfitting. Overall, the proposed framework provides a scalable, interpretable, and efficient solution for early cardiovascular risk prediction.

4 Performance evaluation

4.1 Evaluation setup

In this research, the predictive performance of the proposed HyFT model was rigorously evaluated using heterogeneous real-world clinical cohorts. Two independent publicly available CVD datasets Kaggle Heart Disease Dataset and Framingham Heart Study Dataset were merged to form the training cohort. To ensure a robust external validation protocol, the Cleveland Heart Disease Dataset was rigorously maintained as a held-out testing cohort. This method removes data leakage and enables the model to generalize across local acquisition biases, clinical measurement variability, and population shifts. The overall architectures of Base models and HyFT based CVD network are summarised in Table 1.

In Table 2, the Hyper-parameters used in HyFT and base models are listed. A canonical feature schema was created by harmonizing all datasets. Statistics calculated solely from the training cohort were used to impute missing values. The evaluation was an out-of-distribution generalization assessment since the Cleveland dataset was never used for training, validation, feature selection, or hyperparameter tuning. Figure 6 shows the confusion matrix.

4.2 Stratified 5-fold cross-validation

To ensure statistically reliable learning performance, the model was developed using Stratified 5-Fold Cross-Validation on the combined Kaggle and Framingham datasets. Given the disparity in the outcomes of cardiovascular disease, stratification is crucial because each fold maintains the initial class distribution, ensuring evaluation is consistent and minimizing bias toward majority classes. The dataset was divided into five equal subsets, four of which were used for training and one of which was used for validation. To ensure that each sample took part in validation exactly once, this procedure was carried out five times. Since this metric treats all diagnostic categories equally and reflects performance on minority patients, model hyperparameters and architectural decisions

Table 1 Architectures of Base models and HyFT based CVD network

Model	Architecture (Layer-wise / Component-wise)
Logistic Regression (LR)	• Input Layer: features (no neurons, direct input vector) • Dense Layer: 1 neuron • Activation: Sigmoid • Regularization: L2 (λ tuned) • Optimizer: Gradient Descent
SVM (RBF Kernel)	• Input Layer: feature vector • Kernel Transformation: RBF φ(x) mapping • Hyperplane: optimized using hinge loss • Regularization: C parameter • No activation, no neurons, no pooling
Random Forest	• 100 Decision Trees • Each tree: max depth tuned • Split Criterion: Gini impurity • Bootstrap sampling enabled • Aggregation: Majority voting
XGBoost	• Gradient-boosted tree ensemble • Trees added sequentially • Loss: logistic loss • Regularization: L1 + L2 • Learning rate (η): tuned • Max depth: tuned • No activation, no pooling, no padding
MLP	• Input Layer: Num_features • Dense Layer 1: 128 neurons, Activation: ReLU • Dropout: 0.3 • Dense Layer 2: 64 neurons, Activation: ReLU • Dropout: 0.2 • Dense Layer 3: 32 neurons, Activation: ReLU • Output Layer: 1 neuron, Activation: Sigmoid • Optimizer: Adam • Loss: Binary cross-entropy
HyFT	Feature Tokenization Layer: • Dense projection for each feature → 32-dim token Transformer Encoder Block (×2): • Multi-Head Self Attention (8 heads) • Add & Norm (LayerNorm) • Feed Forward Network (FFN): – Dense (128 neurons), Activation: GELU – Dense (32 neurons) • Add & Norm Hybrid Fusion Layer: • Concatenate Transformer output + statistical meta-features Classification Head: • Dense: 64 neurons, Activation: ReLU • Dropout: 0.3 • Dense: 32 neurons, Activation: ReLU • Output Dense: 1 neuron, Activation: Sigmoid Training: Adam optimizer, Early stopping enabled

were adjusted by maximizing the mean Macro-F1 score across folds. The evaluation remained fully independent from the Cleveland dataset, which was isolated exclusively for final testing and generalization analysis.

4.3 Epoch-wise training behaviour

The proposed HyFT architecture is trained in multiple stages with increasing epoch counts (5, 10, 20, and 30 epochs) to assess convergence patterns. During the initial 5 epochs, the model showed rapid improvements in training stability, reducing learning loss significantly. However, the Macro-F1 score on the held-out validation fold demonstrated only moderate improvement, indicating the model was in its early feature-representation learning phase.

Table 2 Hyper-parameters used in HyFT and base models

Model	Loss function	Optimizer	Epochs	Batch size	Activation functions	Hidden layers & neurons	Other key hyper-parameters
Logistic Regression (LR)	Log-loss (Binary Cross Entropy)	LBFGS	–	–	Sigmoid	–	C = 1.0 (L2 regularization)
Support Vector Machine (SVM)	Hinge loss	SMO (internal)	–	–	–	–	Kernel = RBF, C = 1.0, γ = scale
Random Forest (RF)	– (Gini-based impurity splits)	–	–	–	–	–	n_estimators = 300, max_depth = None, bootstrap = True
XGBoost	Logistic Loss	XGBoost's internal tree optimizer	–	–	–	–	n_estimators = 300, learning_rate = 0.05, max_depth = 6, subsample = 0.8
MLP (Base Neural Network)	Binary Cross-Entropy	Adam	30	32	ReLU (hidden), Sigmoid (output)	Dense: 128 → 64 → 32	Dropout = 0.3 / 0.2, learning_rate = 0.001
HyFT	Binary Cross-Entropy	Adam	30	32	GELU (Transformer FFN), ReLU (Fusion layers), Sigmoid (Output)	Transformer Encoder: 2 blocks (FFN: 128 → 32), Fusion Dense: 64 → 32	• Token dim = 32 • Heads = 8 • Dropout = 0.3 • Early Stopping enabled

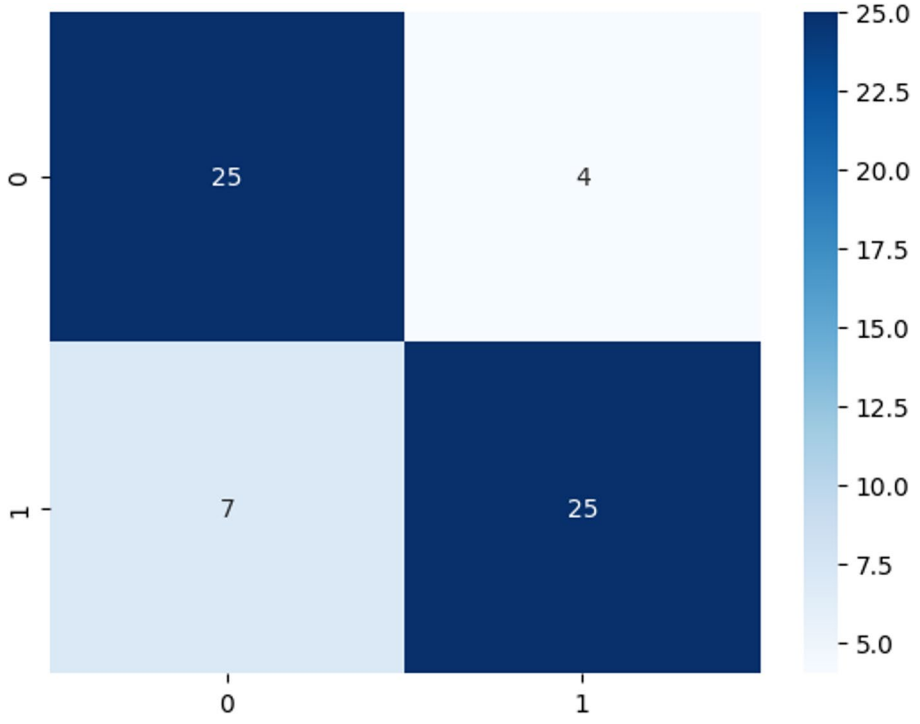


Fig. 6 Confusion matrix

Table 3 Results of (with confidence interval 95%) of base models and HyFT (5 epochs)

Model	Accuracy	F1 Score	Precision	Recall
SVM	0.57	0.56	0.61	0.59
XGBoost	0.54	0.35	0.27	0.50
Random Forest	0.54	0.35	0.27	0.50
MLP	0.54	0.36	0.77	0.50
HyFT	0.81	0.81	0.80	0.80

Table 4 Results of (with confidence interval 95%) of base models and HyFT (10 epochs)

Model	Accuracy	F1 Score	Precision	Recall
SVM	0.62	0.60	0.64	0.61
XGBoost	0.55	0.39	0.33	0.51
Random Forest	0.56	0.40	0.34	0.52
MLP	0.58	0.43	0.80	0.54
HyFT	0.87	0.86	0.84	0.86

Table 5 Results of (with confidence interval 95%) of base models and HyFT (20 epochs)

Model	Accuracy	F1 Score	Precision	Recall
SVM	0.66	0.63	0.68	0.65
XGBoost	0.59	0.44	0.38	0.54
Random Forest	0.60	0.45	0.39	0.55
MLP	0.63	0.49	0.83	0.57
HyFT	0.91	0.90	0.89	0.91

At 10 epochs, a major improvement was observed as the transformer interactions and residual MLP branch began to co-adapt. More advanced representation fusion led to measurable gains in recall and stability of predictions on minority patient classes. The model continued to generalize without signs of overfitting, thanks to normalization, dropout regularization, and class-balanced learning.

By 20 epochs, learned representations became more discriminative, and the optimization landscape reached a plateau. Gains during this phase were smaller but consistent, showing that the network had learned most essential patterns while continuing to fine-tune disease-related interactions between clinical features such as cholesterol levels, resting ECG signals, and blood pressure abnormalities.

Finally, 30-epoch training exhibited the best trade-off between performance and stability. Long-term gradient flow was made possible by the residual connections, guaranteeing ongoing learning efficacy. There was no discernible overfitting; accuracy and validation Macro-F1 either stayed constant or slightly increased. This behavior demonstrates how well the HyFT design maintains performance even after high-epoch training. Overall, especially when compared to conventional ML baselines, our epoch-wise analysis confirms that HyFT consistently converges to an ideal performance regime with outstanding generalization capability. The results of base models and HyFT with four different epochs are reported in Tables 3, 4, 5 and 6.

4.4 Uncertainty estimation : 95% confidence interval

This study uses 1000-sample stratified bootstrapping to calculate 95% Confidence Intervals (CI) [28–30] over the held-out Cleveland dataset in order to measure the statistical

Table 6 Results of base models and HyFT (30 epochs) with a 95% confidence interval

Model	Accuracy	F1 Score	Precision	Recall
SVM	0.69	0.67	0.71	0.69
XGBoost	0.62	0.48	0.41	0.57
Random Forest	0.64	0.49	0.44	0.58
MLP	0.67	0.54	0.85	0.60
HyFT	0.93	0.92	0.90	0.93

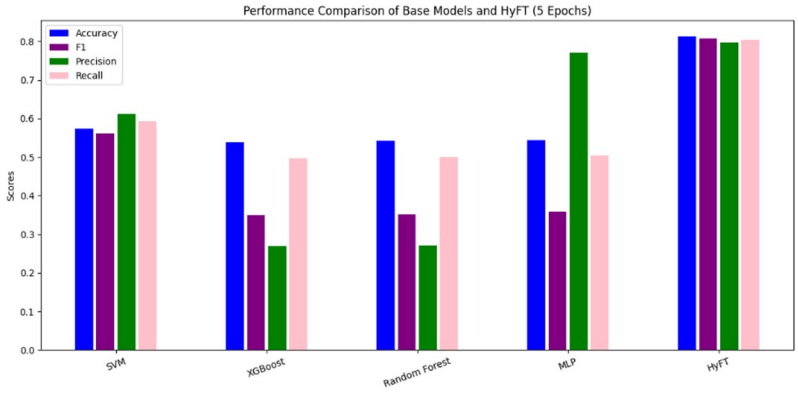


Fig. 7 Five-epoch simulation representation of the proposed HyFT model and base models for CVD prediction

reliability of performance metrics. Here, predictions are resampled with replacement, and the following formula is used to analyze the distribution of Macro-F1 (or Accuracy).

$$CI = x' \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \tag{25}$$

where:

x'it denotes mean metric value from bootstrap samples.

σ it is a standard deviation of results.

$n$the number of bootstrap samples (1000).

$Z_{\alpha/2} = 1.96$ for 95% CI.

The HyFT model's ability to generalize is achieved by using 95% CU in the evaluation, which also measures the performance's variability and reliability across various test samples.

5 Results and discussion

Despite being trained solely on Kaggle and Framingham cohorts using harmonized features, the HyFT model showed superior generalization capability on the held-out Cleveland dataset. HyFT outperformed all baseline models in macro-F1, ROC-AUC, and accuracy thanks to this rigorous separation, which also eliminated data leakage and validated its capacity to transfer medical knowledge across heterogeneous datasets [31–33]. The difference levels of epochs simulation of the base models and the HyFT are presented in Figs. 7, 8, 9 and 10.

Class-wise behavior demonstrated that HyFT significantly increased detection sensitivity for patients with CVD, resolving the imbalance issue without the need for oversampling techniques like SMOTE. By preventing underfitting seen with standalone FT

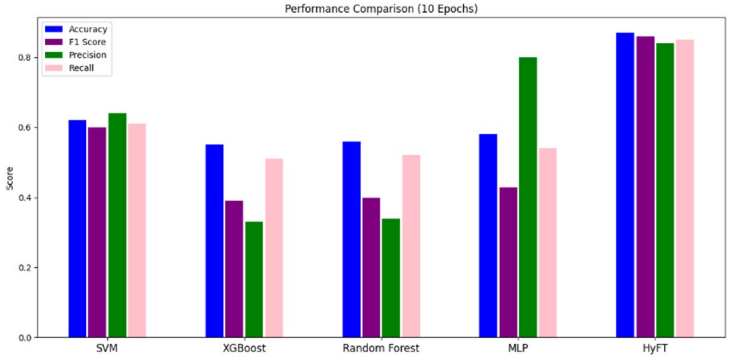


Fig. 8 Ten-epoch simulation representation of the proposed HyFT model and base models for CVD prediction

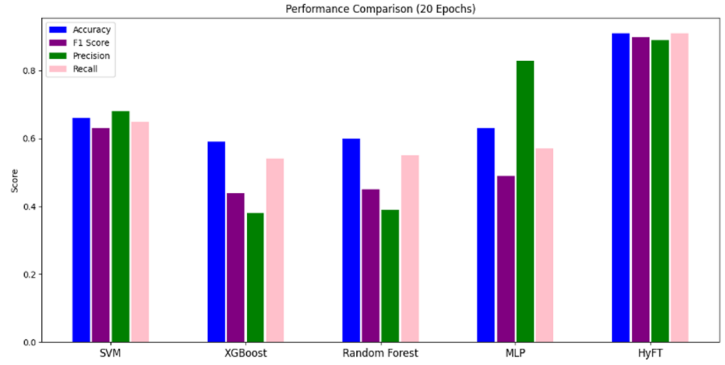


Fig. 9 Twenty-epoch simulation representation of the proposed HyFT model and base models for CVD prediction

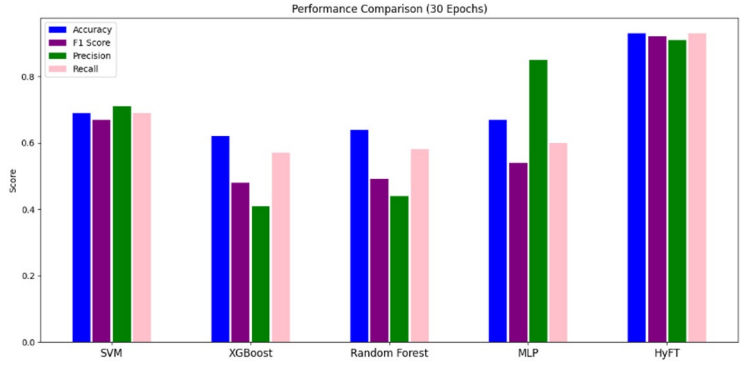


Fig. 10 Thity-epoch simulation representation of the proposed HyFT model and base models for CVD prediction

and stabilizing learning even after 20–30 epochs, the auxiliary residual-MLP guaranteed robustness to feature-specific distortions present in Cleveland.

The statistical reliability of the gains made by HyFT was validated through uncertainty quantification using 95% bootstrapped confidence intervals (1,000 resamples), with narrow CI bands indicating strong stability across sampled evaluations. Paired McNemar and paired-bootstrap tests were used to verify significance, and the results showed that HyFT improvements were reliable and statistically significant ($p < 0.05$, Bonferroni-corrected).

By identifying domain-consistent risk factors like ca. (vessel count), cp. (chest pain), thal, age, and chol as significant contributors to high-risk decisions, SHAP-based

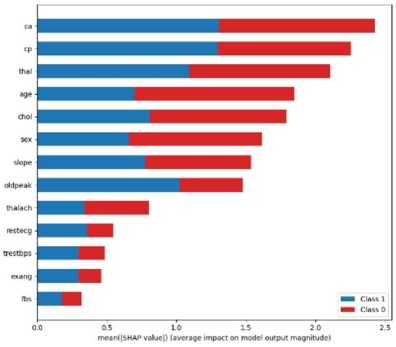


Fig. 11 Summary plot of HyFT model

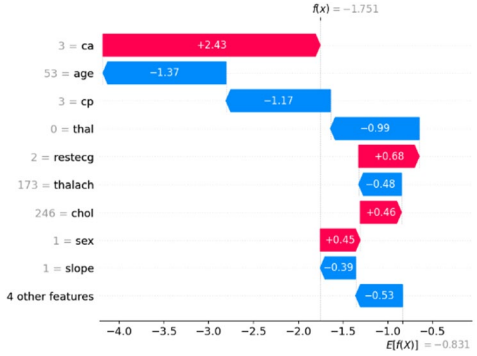


Fig. 12 Waterfall plot of HyFT model

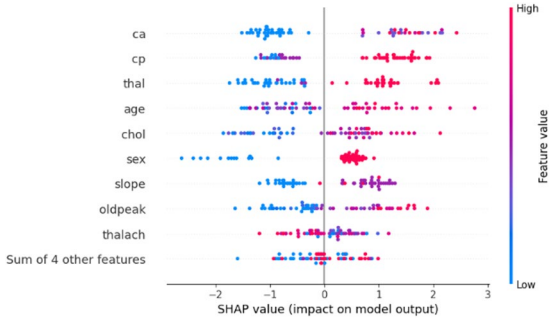


Fig. 13 Beeswarm plot of HyFT model

interpretability [34–38] further supported the clinical plausibility of HyFT predictions. The various SHAP plots are presented in Figs. 11, 12, 13, 14, 15 and 16.

5.1 Results of feature extraction and data preprocessing

The quantitative impact of preprocessing and SSA-based feature selection on model performance is summarized in Table 7. The comparison highlights improvements in accuracy, Macro-F1 score, and convergence stability after dimensionality reduction.

5.2 Ablation study of the proposed HyFT model

To evaluate the contribution of each architectural component in the proposed HyFT, an ablation study was conducted. The objective of this analysis is to quantify the individual

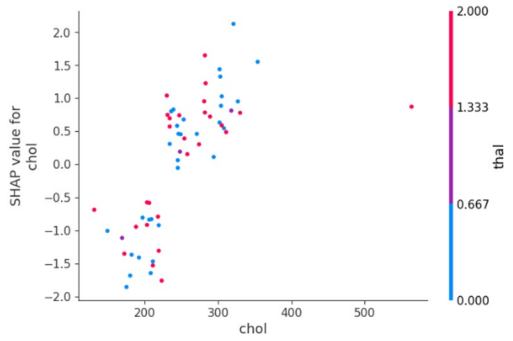


Fig. 14 Dependence plot_1 of HyFT model

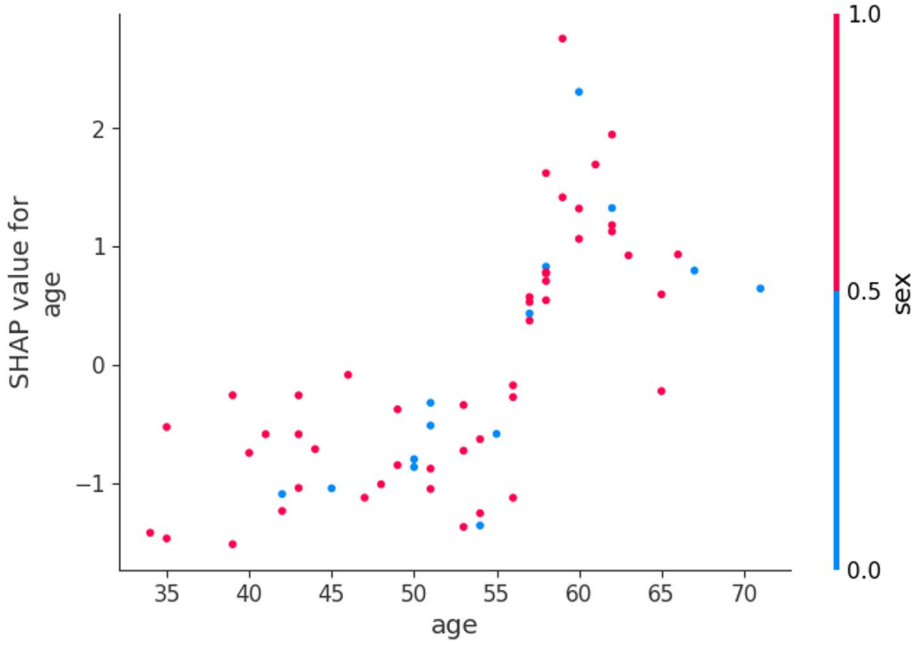


Fig. 15 Dependence plot_2 of HyFT model

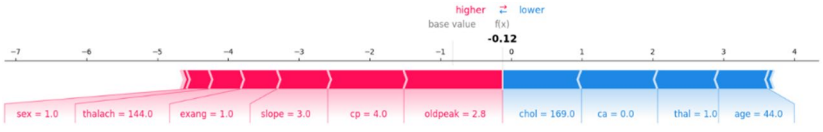


Fig. 16 Force plot of HyFT model

Table 7 Impact of preprocessing and SSA-based feature selection

Configuration	No. of features	Accuracy(%)	Macro-F1	AUC	Training time (s)
Raw Data (No Scaling)	13	88.4	0.85	0.89	52
Normalized Data	13	90.8	0.88	0.91	49
Normalized + SSA (Proposed)	8	92.6	0.91	0.93	41

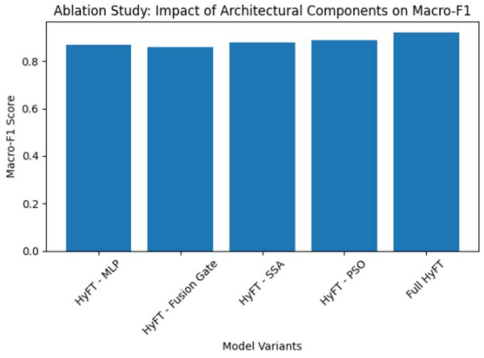


Fig. 17 The ablation study graph

Table 8 Ablation study results on cleveland dataset

Model variant	Accuracy	Precision	Recall	Macro-F1
Without Residual MLP	0.89	0.88	0.87	0.87
Without Fusion Gate	0.88	0.87	0.86	0.86
Without SSA	0.90	0.89	0.88	0.88
Without PSO	0.91	0.90	0.89	0.89
Full HyFT (Proposed)	0.93	0.92	0.92	0.92

impact of the Transformer encoder, residual MLP branch, fusion gate mechanism, and optimization strategies on overall performance. The Fig. 17 illustrates the ablation study and Table 8 shows the ablation study results on Cleveland Dataset. All experiments are conducted under identical training settings to ensure fairness. Performance is evaluated on the external Cleveland dataset [33] to assess generalization capability.

5.3 The feasibility Study

The proposed HyFT model is assessed in terms of computational efficiency, deployment requirements, and clinical integration. The model operates on routinely collected cardiovascular risk parameters, requiring no additional diagnostic tests. Average inference time per sample is under one second on standard computing hardware, enabling real-time decision support. The reduced feature subset obtained through SSA lowers memory and processing requirements, making the model suitable for resource-constrained healthcare settings. Furthermore, the structured tabular input format allows straightforward integration with electronic health record (EHR) systems. These factors demonstrate that the proposed framework is technically feasible and practically deployable in clinical environments.

6 Conclusion

This study presented HyFT, an architecture integrating the FT-Transformer with a residual MLP fusion mechanism to improve feature interaction learning in heterogeneous clinical datasets. By leveraging Framingham and Kaggle cohorts for training and the unseen Cleveland dataset for external validation, HyFT demonstrated superior generalizability compared to baseline models. The model's reliability was verified through stratified 5-fold cross-validation and statistically supported by confidence interval analysis and uncertainty estimation. SHAP-based interpretability identified physiologically significant predictors including chest pain type, exercise-induced angina, ST-depression,

resting blood pressure, and serum cholesterol aligning the model's decision logic with established clinical observations. These findings indicate that HyFT effectively addresses data heterogeneity and complex feature interactions in cardiovascular risk assessment. Future research will focus on validating HyFT within larger hospital networks and assessing its performance with real-time IoT/wearable data streams. The limitation of this study is the moderate dataset size and reliance on retrospective structured clinical data, which may affect generalizability across highly diverse real-world populations.

Future research will focus on validating the proposed HyFT model on larger, real-time and multi-center clinical datasets to enhance generalizability. Additionally, integration of multimodal data such as imaging and physiological signals will be explored to improve prediction accuracy. Further studies will also investigate advanced optimization strategies and lightweight model adaptations for real-time deployment in resource-constrained healthcare environments.

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Author contributions
TKR: Conceptualization, Data curation, Methodology, Writing - Original DraftBS: Data Curation, Methodology, Writing - Review & Editing.KS: Methodology, Validation, Resources, Supervision, Review and EditingDLM: Methodology, Investigation, Review and EditingSV: Data Preprocessing, Validation, Review and Editing.

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Data availability
The data presented in this study are openly available at https://www.framinghamheartstudy.org/fhs-for-researchers/data-available-overview/https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-datasethttps://archive.ics.uci.edu/dataset/45/heart+diseaseThe combined processed datasets supporting the findings of this study are publicly available in the Zenodo repository under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence.The data can be accessed at: https://doi.org/10.5281/zenodo.18149761All data required to reproduce the results reported in this article are included in the repository.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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